ASHUTOSH NITINKUMAR LATWALA

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Professional Summary

Motivated and result-driven Software Engineer and AI Engineer with 10+ years delivering data-driven applications that blend modern software engineering with practical AI and analytics. Strong in Python, Java, cloud platforms (including AWS, GCP), and data analytics, with a track record of building scalable, secure, and maintainable systems from ingestion and ETL through APIs and front-end delivery. Experienced implementing machine learning solutions, optimizing data pipelines, and shipping end-to-end products that improve performance, reduce cost, and accelerate decision making. Recent work includes building analytics platforms and retrieval-augmented experiences with LLMs/SLMs, schema-constrained APIs, and configurable navigation systems that improve user experience and developer velocity. Known for disciplined execution in Agile teams, and turning complex requirements into measurable, production-ready outcomes.

Education

Master of Science in Information Systems (DS & AI Track) — **January 2023 – December 2024**
University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA — **GPA:** 3.97/4.0

Bachelor of Engineering in Computer Science & Engineering — **June 2015 – May 2019**
C. K. Pithawala College of Engineering & Technology — **GPA:** 3.97/4.0
Gujarat Technological University, Ahmedabad, Gujarat, India

Skills

Languages	Python, Java, JavaScript, SQL, PL/SQL, C, C++
Backend	Java (J2EE), Spring Boot, Spring MVC, Django, RESTful APIs, SOAP, Microservices, Web Development, Deployment & Maintenance, PHP, Node.js, Flask, FastAPI
Frontend	Angular, Bootstrap, CSS, HTML, Angular Router, Angular Navigation
Databases	MySQL, SQL Server, MongoDB, PL/SQL, PostgreSQL, Oracle, DB2
Cloud / DevOps	AWS (EC2, Lambda, S3, RDS, Bedrock, CloudWatch, Secret Manager, SES, Cognito, IAM), Google Cloud Platform (GCP) Services, ETL, CI/CD
AI / ML & Data	Data Structures, Data Mining, AI/ML, LLMs, SLMs, RAG, Generative AI, Prompt Engineering, Data Analysis, Data Analytics, Data Visualization
Testing / Logging	BDD, JUnit, Mockito, Cucumber, Gherkin, Selenium, Postman, Log Analysis, Splunk
Other Tools	Git, GitHub, Bitbucket, Docker, Tableau, Power BI, MS Office, BigCharts, Polygon, Hugging Face, HubSpot, Asana, Jira, Confluence, Plaid, D-ID, VS Code

Professional Experience

AI Engineer (IT) — **June 2024 – Present**
Center for Applied AI, UMBC Training Centers. — Columbia, MD

- Developed Quantiv.ai as a full-stack analytics platform using Flask/FastAPI, Angular, and AWS (EC2, S3, RDS, CloudWatch, Cognito, SES) following SCRUM ceremonies and iterative demos.
- Designed and implemented RESTful microservices with OpenAPI, enforced OAuth2/Cognito authentication, RBAC, and token lifecycle management.
- Engineered ETL pipelines to ingest market/option data, compute KPIs, and persist backtest results for longitudinal analysis and reporting.
- Integrated Polygon.io market feeds, built asynchronous workers with queued processing, retry/backoff, and rate-limit governance.
- Contributed to AI research and integration, evaluating AI assistants for team productivity and enhanced the C4AI.ai website through development and testing.
- Optimized AI systems by monitoring performance, conducting data-driven analysis, and applying cutting-edge methodologies for continuous improvement.
- Implemented Angular Router navigation (lazy loading, guards, deep-linking, breadcrumbs) with config-driven menus reused across features.
- Built responsive HTML/CSS/Bootstrap components and shared UI modules to standardize UX and reduce front-end defects.
- Connected D-ID for AI video explainers, automated media storage and delivery via S3 pipelines and SES email

workflows.

- Produced LLM-ready APIs with JSON-safe structured outputs for downstream retrieval/summarization and auditability.
- Extended backend development in Python (Flask/FastAPI) with async workers, typed DTOs, and pandas utilities for robust data transforms and validation.
- Added MongoDB for schemaless event logging and caching of transient job metadata; implemented TTL indexes and aggregation pipelines for cleanup and reporting.
- Integrated legacy data sources in DB2, built ETL stages to normalize schemas and reconcile records with RDS datasets.
- Built an LLMs/SLMs with RAG, document chunking, hybrid retrieval, and JSON-schema-constrained outputs for reliable, auditable responses.
- Adopted Hugging Face embeddings to pilot LLM/SLM search and RAG-style FAQ responses for help content.
- Leveraged Hugging Face models/embeddings and tokenizer utilities, packaged artifacts with versioned model cards and reproducible inference configs.
- Formalized Git workflows using Git/GitHub, status checks in CI for higher merge quality.
- Created a config-driven Angular navigation framework (role-aware menus, route guards, lazy modules) and shared UI configuration reused across multiple functionalities.
- Tuned SQL queries, added indexes, and introduced server-side pagination/caching to reduce latencies on RDS.
- Centralized logging/metrics/traces with CloudWatch dashboards/alarms and actionable runbooks.
- Established CI/CD with GitHub Actions, enabled blue/green deployments and environment promotion.
- Secured secrets with AWS Secrets Manager, tightened security headers/CORS and applied OWASP input validation and request throttling.
- Containerized services with Docker and parameterized builds for dev/stage/prod parity, documented image hardening steps.
- Authored architecture diagrams, sequence/API handbooks, and operational playbooks, maintained artifacts in Jira/Confluence.
- Practiced BDD using Cucumber (Gherkin feature files), extended automated regression with Selenium and Postman/Newman collections.
- Implemented WebSocket for long-running jobs to improve user feedback and perceived responsiveness.
- Exposed health/actuator endpoints with liveness/readiness probes to support rolling updates and auto-recovery.
- Conducted capacity tests and profiling, right-sized instances and tuned thread/connection pools to balance cost, throughput, latency.
- Coordinated cross-functional grooming and sprint planning, refined epics into stories, demoed increments, and captured acceptance criteria.
- Implemented error budgets and SLOs for key endpoints, instrumented alerts on saturation, errors, and tail latency.
- Built Postman contract tests and mock servers to unblock Angular teams during backend changes and versioned API transitions.
- Mentored peers on API design, Angular navigation patterns, and secure coding practices; contributed code reviews and standards.
- Documented system architecture, API flows, and user behavior, aiding internal handoff and future development/maintenance.
- Researched and tested Plaid API for financial data aggregation, enabling secure access to user bank accounts, transaction history, and account metadata.
- Built and validated custom Postman test suites to explore Plaid endpoints, simulate user account linkage, and extract key financial data using request/response scripting.
- Developed logic in Postman to filter and extract account names, balances, and liabilities based on specific user attributes, using chained requests and JavaScript tests.
- Collaborated with team members and attended regular meetings, ensuring smooth project execution and alignment with organizational goals.
- Partnered with managers and clients to gather business requirements, translate them into system specifications, and assist in business/IT solutions implementation.

Associate Software Engineer **August 2019 – December 2022**
Duma Infoservices Pvt. Ltd. Surat, Gujarat, India

- Practiced Agile/SCRUM delivery: planned sprints, ran stand-ups, and demoed finished stories each iteration to gather actionable user feedback.
- Designed and delivered a full-stack grocery e-commerce platform using Node.js and MongoDB, coordinating sprints, code reviews, and on-time releases.
- Stood up Django backends with REST endpoints for catalog, inventory, and orders, normalized MySQL schemas to improve update speed and reporting fidelity.

- Implemented Spring Boot/Spring MVC services with controller → service → repository layering, added SOAP integrations for a legacy.
- Built Python services with Django/Flask/FastAPI, defining clean REST endpoints, input validation, and pagination for list views.
- Crafted Angular front ends with a Router-driven navigation system (lazy modules, guards, deep links) and configurable menus consumed from an API.
- Standardized UI with HTML/CSS/Bootstrap design tokens and reusable components, cutting CSS drift and rework.
- Created config-driven navigation and shared UI configuration so menus/permissions could be updated centrally across multiple features.
- Built an ML recommendation service (Python/TensorFlow/Flask) for "similar" and "frequently bought together," exposed via cached REST routes.
- Built ETL pipelines (Python + SQL) to normalize vendor feeds, deduplicate records, and reconcile inventory against POS exports.
- Built ETL tasks in Python/SQL to ingest CSV/JSON feeds, deduplicate records, and reconcile transactional data for downstream reporting.
- Modeled data for PostgreSQL reporting tables and wrote PL/SQL procedures; tuned indexes after analyzing query plans.
- Designed PostgreSQL schemas, wrote complex SQL/PL/pgSQL for reports and batch jobs, and enforced integrity with constraints.
- Implemented DB2 stored procedures and bulk-load jobs, coordinated data sync tasks with SQL Server and Oracle using scheduled scripts.
- Integrated legacy sources using DB2/Oracle/SQL Server connectors, scripted migrations and automated validation checks.
- Practiced BDD with Cucumber (Gherkin), pairing with JUnit/Mockito for Java modules and PyTest for Python services.
- Containerized workloads with Docker and implemented CI/CD on Git/GitHub/Bitbucket for reproducible builds and environment promotions.
- Deployed workloads on AWS EC2/RDS/S3 with CloudWatch alarms, log groups, and dashboards, used GCP Cloud Storage/Build for select utilities.
- Built small utility APIs in Flask and FastAPI for internal tools (file processing, reporting).
- Packaged apps for deployment with Gunicorn behind Nginx, added environment-based configs, and scripted release steps for repeatability.
- Containerized services with Docker (multi-stage builds), standardized base images, and ensured parity.
- Secured endpoints with Spring Security (form login, RBAC) and standardized error payloads.
- Performed data analytics with Pandas/SQL, computing KPIs and publishing weekly summaries.
- Created interactive Tableau dashboards (filters, parameters, calculated fields) to visualize trends, exceptions, and SLA adherence for stakeholders.
- Ran Splunk queries and dashboards for error budgets and latency trends.
- Authored runbooks, sequence diagrams, and deployment notes in Confluence, kept the backlog healthy in Jira.
- Established branch strategy and PR templates in GitHub/Bitbucket, enabled required checks and CODEOWNERS for safer merges.
- Orchestrated web deployment & maintenance activities: health checks, blue/green rollouts, and feature-flagged launches.
- Collaborated on full project lifecycles, from design to deployment, ensuring high-quality deliverables and enhancing team efficiency, while constantly improving skills in Python, web technologies, and cloud platforms.
- Fostered effective team collaboration through regular meetings, ensuring seamless project execution and alignment with strategic objectives.
- Conducted data collection and analysis, utilizing various methods including interviews, surveys, and database search, to identify trends and provide reports for business development.
- Implemented unit and integration testing using Postman and PyTest, ensuring reliability and reducing bugs by 30% before production deployment.
- Contributed to the automation of deployment processes using Git, streamlining CI/CD workflows and minimizing manual intervention.

Software Developer **June 2015 – July 2019**
Narola Infotech Surat, Gujarat, India

- Developed the application using Agile and SCRUM methodology, participated in Scrum meetings where we would demo all the user stories done during that iteration for final feedback from end users.
- Developed enterprise web apps in Java (J2EE), Spring MVC/Spring Boot, and REST/SOAP with layered DTO/DAO architecture.

- Built Angular SPAs with reusable components, Router-based navigation (lazy loading, guards), and role-aware menus.
- Built responsive webpages with HTML/CSS/Bootstrap components.
- Implemented data access with Hibernate/JDBC, wrote complex SQL/PLSQL and tuned queries for performance.
- Integrated multiple databases (DB2, Oracle, MySQL, SQL Server) and abstracted persistence for portability.
- Designed ETL jobs using Spring Batch for nightly imports/exports with retry and audit tables.
- Set up CI/CD (Jenkins/Git), including static analysis, JUnit/Mockito unit tests, and Selenium smoke tests.
- Deployed to AWS EC2 and on-prem, configured Nginx/Apache, SSL/TLS, and health checks.
- Instrumented observability with Splunk dashboards and alert thresholds to reduce incident response time.
- Implemented Spring Security with form/OAuth2 login, RBAC, and method-level authorization.
- Introduced caching and server-side pagination, reduced DB load and improved response times for high-traffic endpoints.
- Created Postman/Newman suites for API regression and release validation across environments.
- Implemented centralized exception handling and uniform error payloads for traceability across services.
- Wrote functional/technical specs and maintained versioned docs in Confluence, enforced PR reviews in Git/Bitbucket.
- Built file processing pipelines, checksum validation, and quarantine flows for malformed data.
- Optimized JVM settings, connection pools, and thread pools to stabilize high-throughput services.
- Practiced Agile delivery with two-week sprints, visible burn-down, and definition-of-done checklists.
- Coordinated data migrations and schema versioning; validated outcomes with backfill scripts and reports.
- Supported Web Deployment & Maintenance, including zero-downtime rollouts and fast rollback plans.
- Built scheduled maintenance pages and feature flags to minimize user impact during releases.
- Collaborated in Agile/Scrum to refine requirements, estimate stories, manage sprint scope, and demo increments to stakeholders.
- Collaborated with team members and attended regular meetings, ensuring smooth project execution and alignment with organizational goals.

Additional Experience

Graduate Teaching Assistant **January 2024 – December 2024**

Department of Information Systems, Academic Success Center, UMBC. Baltimore, MD

- Contributed to the quality of student learning in Computer Science, Computer Engineering and Information Systems classes. Provided grading, personalized feedback, proctored exams and facilitated demo sessions.

Projects

Quantiv.ai – Full-Stack AI-Powered Stock & Options Analysis Platform **November 2024 – June 2025**

- Built a secure, full-featured AI powered web application for stock backtesting and options performance analysis using Python, Flask, Chart.js, and AWS services (EC2, S3, Cognito, S3, Secret Manager, SES).
- Integrated Polygon.io for real-time financial data and D-ID API for AI-generated video explanations.
- Enabled users to perform strategy-based backtests, generate explainer videos, record sessions, S3 upload and share insights via email, all through a seamless UI with authentication and data protection.

Published Research: Correlated Anomalous Pattern Mining from Cybersecurity Transactional Datasets
September 2024 – December 2024

- Developed a novel data mining algorithm using FP-Growth and correlation analysis to detect correlated anomalous patterns in cybersecurity logs, leveraging the NSL-KDD dataset.
- Focused on identifying infrequent but significant patterns that co-occur in attack scenarios using anti-monotone and strong affinity (cosine and Jaccard) properties.
- Demonstrated early threat detection potential through correlation between network features (e.g., protocol, service, flags) and known attacks (e.g., Neptune, Smurf).
- Evaluated algorithm performance with complexity analysis, runtime, and memory metrics, showing scalability across both training and test subsets.

Experiment & Research Paper on **January 2023 – May 2023**

- Encryption: "Essential for data security and privacy in digital age, but its effectiveness is threatened by the advances in computing power, new technologies and the emergence of new threats to cybersecurity."
- Explored how modern encryption algorithms withstand evolving threats, including quantum computing.
- Conducted experiments comparing symmetric and asymmetric encryption in terms of performance, complexity, and vulnerability exposure.

Impact of Covid-19 on IT Business **August 2020 – April 2021**

- Developed Python based IT business analysis model using ML algorithms to predict the IT business success

during Covid-19 pandemic by considering related factors.
- Achieved over 85% accuracy in predicting business performance using decision tree and logistic regression models on historical datasets.

Smart Street Light **October 2019 – March 2020**

- Designed a system using hardware software infrastructure and Arduino-uno, in which the light intensity changes according to traffic density to save electricity.
- Integrated IR sensors and real-time microcontroller programming to optimize energy usage and enhance automation efficiency.

Training & Certification

- Oracle Cloud Data Management 2023 Foundation Associates, Oracle **July 2024**
- Python, Java Programming **October 2023**
- HTML, CSS & JavaScript for Web Developers Course, Coursera **July 2021**
- Artificial Intelligence Classroom Series, Microsoft **March 2021**
- Certificate of Excellence Top Ranker 2018, 2019, GTU **July 2019, July 2018**
- Certification in Digital Wellness and Cyber Security **February 2018**